

## Modal Logic: More Tableaux Examples

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2026-02-23

## Midterm

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- The exam will be eight questions long.
- The questions will be like the ones on this week's quiz and the practice exam.
- It will be here, with answers to be done on paper.
- We'll find a separate room (to be announced) for folks needing accommodations.
- You'll get the cheat sheet, but otherwise it is closed book.

## Quiz 6

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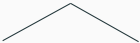
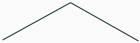
## Question 1

$\Box(A \vee B) \rightarrow (\Diamond A \vee \Box B)$  is a theorem of KT

|                           |                                                                       |                            |
|---------------------------|-----------------------------------------------------------------------|----------------------------|
| 1.                        | $\mathbb{F} 1, \Box(A \vee B) \rightarrow (\Diamond A \vee \Box B)$ ✓ | Assumption                 |
| 2.                        | $\top 1, \Box(A \vee B)$                                              | $\rightarrow \mathbb{F} 1$ |
| 3.                        | $\mathbb{F} 1, \Diamond A \vee \Box B$ ✓                              | $\rightarrow \mathbb{F} 1$ |
| 4.                        | $\mathbb{F} 1, \Diamond A$                                            | $\vee \mathbb{F} 3$        |
| 5.                        | $\mathbb{F} 1, \Box B$ ✓                                              | $\vee \mathbb{F} 1$        |
| 6.                        | $\mathbb{F} 1.1, B$                                                   | $\Box \mathbb{F} 5$        |
| 7.                        | $\top 1.1, A \vee B$ ✓                                                | $\Box \top 2$              |
| $\swarrow \quad \searrow$ |                                                                       |                            |
| 8.                        | $\top 1.1, A \quad \top 1.1, B$                                       | $\vee \top 7$              |
| 9.                        | $\mathbb{F} 1.1, A \quad \otimes$                                     | $\Diamond \mathbb{F} 4$    |
|                           | $\otimes$                                                             |                            |

## Question 2

$\Box(A \rightarrow B) \rightarrow (A \rightarrow \Box B)$  is not a theorem of KT

|                                                                                   |                                                                            |                            |
|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------|----------------------------|
| 1.                                                                                | $\mathbb{F} 1, \Box(A \rightarrow B) \rightarrow (A \rightarrow \Box B)$ ✓ | Assumption                 |
| 2.                                                                                | $\mathbb{T} 1, \Box(A \rightarrow B)$                                      | $\rightarrow \mathbb{F} 1$ |
| 3.                                                                                | $\mathbb{F} 1, A \rightarrow \Box B$ ✓                                     | $\rightarrow \mathbb{F} 1$ |
| 4.                                                                                | $\mathbb{T} 1, A$                                                          | $\rightarrow \mathbb{F} 3$ |
| 5.                                                                                | $\mathbb{F} 1, \Box B$                                                     | $\rightarrow \mathbb{F} 3$ |
| 6.                                                                                | $\mathbb{T} 1, A \rightarrow B$ ✓                                          | $\mathbb{T} \Box 2$        |
|  |                                                                            |                            |
| 7.                                                                                | $\mathbb{F} 1, A$ $\mathbb{T} 1, B$                                        | $\rightarrow \mathbb{T} 6$ |
| 8.                                                                                | ⊗ $\mathbb{F} 1.1, B$                                                      | $\Box \mathbb{F} 5$        |
| 9.                                                                                | $\mathbb{T} 1.1, A \rightarrow B$ ✓                                        | $\Box \mathbb{T} 2$        |
|  |                                                                            |                            |
| 10.                                                                               | $\mathbb{F} 1.1, A$ $\mathbb{T} 1.1, B$                                    | $\rightarrow \mathbb{T} 9$ |
|                                                                                   | ⊗                                                                          |                            |



## Question 2

The middle branch stays open, and the open-ended instruction at line 2 has been applied everywhere that's needed.

The counterexample is where  $A$  and  $B$  are both true at  $w_1$  and both false at  $w_{1.1}$ .

### Question 3

$\Box A \rightarrow (A \wedge \Box \Box A)$  is not a theorem of K4

|                           |                                                             |                            |
|---------------------------|-------------------------------------------------------------|----------------------------|
| 1.                        | $\mathbb{F} 1, \Box A \rightarrow (A \wedge \Box \Box A)$ ✓ | Assumption                 |
| 2.                        | $\mathbb{T} 1, \Box A$                                      | $\rightarrow \mathbb{F} 1$ |
| 3.                        | $\mathbb{F} 1, A \wedge \Box \Box A$ ✓                      | $\rightarrow \mathbb{F} 1$ |
| $\swarrow \quad \searrow$ |                                                             |                            |
| 4.                        | $\mathbb{F} 1, A \quad \mathbb{F} 1, \Box \Box A$ ✓         | $\wedge \mathbb{F} 3$      |
| 5.                        | $\mathbb{F} 1.1, \Box A$                                    | $\Box \mathbb{F} 4$        |
| 6.                        | $\mathbb{T} 1.1, \Box A$                                    | $4 \Box 2$                 |
|                           | ⊗                                                           |                            |

## Question 3

The left hand branch stays open. The counterexample is the somewhat degenerate model where  $w_1$  can't access anything, and has  $A$  false.

## Question 4

$\Diamond(A \rightarrow \Diamond B) \rightarrow (\Box A \rightarrow \Diamond(A \wedge B))$  is a theorem of K4

|                                                                               |                                                                                                            |                            |
|-------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|----------------------------|
| 1.                                                                            | $\mathbb{F} 1, \Diamond(A \rightarrow \Diamond B) \rightarrow (\Box A \rightarrow \Diamond(A \wedge B))$ ✓ | Assumption                 |
| 2.                                                                            | $\mathbb{T} 1, \Diamond(A \rightarrow \Diamond B)$ ✓                                                       | $\rightarrow \mathbb{F} 1$ |
| 3.                                                                            | $\mathbb{F} 1, \Box A \rightarrow \Diamond(A \wedge B)$ ✓                                                  | $\rightarrow \mathbb{F} 1$ |
| 4.                                                                            | $\mathbb{T} 1, \Box A$                                                                                     | $\rightarrow \mathbb{F} 3$ |
| 5.                                                                            | $\mathbb{F} 1, \Diamond(A \wedge B)$                                                                       | $\rightarrow \mathbb{F} 3$ |
| 6.                                                                            | $\mathbb{T} 1.1, A \rightarrow \Diamond B$ ✓                                                               | $\Diamond \mathbb{T} 2$    |
| 7.                                                                            | $\mathbb{T} 1.1, A$                                                                                        | $\Box \mathbb{T} 4$        |
| 8.                                                                            | $\mathbb{F} 1.1, A \wedge B$                                                                               | $\Diamond \mathbb{F} 5$    |
| <div style="text-align: center;"> <math>\swarrow \quad \searrow</math> </div> |                                                                                                            |                            |
| 9.                                                                            | $\mathbb{F} 1.1, A \quad \mathbb{T} 1.1, \Diamond B$ ✓                                                     | $\rightarrow \mathbb{T} 6$ |
| 10.                                                                           | $\otimes \quad \mathbb{T} 1.1, \Box A$                                                                     | $4 \Box 4$                 |
| 11.                                                                           | $\mathbb{F} 1.1, \Diamond(A \wedge B)$                                                                     | $4 \Diamond 5$             |
| 12.                                                                           | $\mathbb{T} 1.1.1, B$                                                                                      | $\Diamond \mathbb{T} 9$    |
| 13.                                                                           | $\mathbb{T} 1.1.1, A$                                                                                      | $\Box \mathbb{T} 10$       |
| 14.                                                                           | $\mathbb{F} 1.1.1, A \wedge B$ ✓                                                                           | $\Diamond \mathbb{F} 11$   |
| <div style="text-align: center;"> <math>\swarrow \quad \searrow</math> </div> |                                                                                                            |                            |
| 15.                                                                           | $\mathbb{F} 1.1.1. A \quad \mathbb{F} 1.1.1. B$                                                            | $\wedge \mathbb{F} 14$     |

$\Diamond(A \wedge B) \rightarrow \Box\Diamond(C \rightarrow B)$  is not a theorem of S4

|     |                                                                                  |                           |
|-----|----------------------------------------------------------------------------------|---------------------------|
| 1.  | $\mathbb{F} 1, \Diamond(A \wedge B) \rightarrow \Box\Diamond(C \rightarrow B)$ ✓ | Assumption                |
| 2.  | $\mathbb{T} 1, \Diamond(A \wedge B)$ ✓                                           | $\rightarrow\mathbb{F} 1$ |
| 3.  | $\mathbb{F} 1, \Box\Diamond(C \rightarrow B)$ ✓                                  | $\rightarrow\mathbb{F} 1$ |
| 4.  | $\mathbb{T} 1.1, A \wedge B$ ✓                                                   | $\Diamond\mathbb{T} 2$    |
| 5.  | $\mathbb{T} 1.1, A$                                                              | $\wedge\mathbb{T} 4$      |
| 6.  | $\mathbb{T} 1.1, B$                                                              | $\wedge\mathbb{T} 4$      |
| 7.  | $\mathbb{F} 1.2, \Diamond(C \rightarrow B)$                                      | $\Box\mathbb{F} 3$        |
| 8.  | $\mathbb{F} 1.2, C \rightarrow B$ ✓                                              | $\mathbb{T}\Diamond 7$    |
| 9.  | $\mathbb{T} 1.2, C$                                                              | $\rightarrow\mathbb{F} 8$ |
| 10. | $\mathbb{F} 1.2, B$                                                              | $\rightarrow\mathbb{F} 8$ |



## Question 5

The trunk stays open, and we've applied the open-ended rule everywhere it needs to be applied.

The countermodel has  $w_1$ , which can access  $w_{1.1}$  and  $w_{1.2}$ . At  $w_{1.1}$  both  $A$  and  $B$  are true. At  $w_{1.2}$ ,  $C$  is true and  $B$  is false.

Any way of filling in the other values will still create a counterexample.

## Question 6

$\Diamond\Diamond A \rightarrow (\Diamond A \wedge \Diamond\Diamond\Diamond A)$  is a theorem of S4

|                            |                                                                                                 |                           |
|----------------------------|-------------------------------------------------------------------------------------------------|---------------------------|
| 1.                         | $\mathbb{F} 1, \Diamond\Diamond A \rightarrow (\Diamond A \wedge \Diamond\Diamond\Diamond A)$ ✓ | Assumption                |
| 2.                         | $\top 1, \Diamond\Diamond A$ ✓                                                                  | $\rightarrow\mathbb{F} 1$ |
| 3.                         | $\mathbb{F} 1, \Diamond A \wedge \Diamond\Diamond\Diamond A$ ✓                                  | $\rightarrow\mathbb{F} 1$ |
| 4.                         | $\top 1.1, \Diamond A$                                                                          | $\Diamond\top 2$          |
| $\swarrow \qquad \searrow$ |                                                                                                 |                           |
| 5.                         | $\mathbb{F} 1, \Diamond A$ $\mathbb{F} 1, \Diamond\Diamond\Diamond A$                           | $\wedge\mathbb{F} 3$      |
| 6.                         | $\mathbb{F} 1.1, \Diamond A$ $\mid$                                                             | $4\Diamond 5$             |
| 7.                         | $\otimes$ $\mathbb{F} 1, \Diamond\Diamond A$                                                    | $\top\Diamond 6$          |
|                            | $\otimes$                                                                                       |                           |

There are a lot of ways to close this one; this is the quickest I found but it's definitely not the only one.

## Question 7

$\Diamond \Box A \rightarrow \Diamond A$  is a theorem of KTB

|    |                                                          |                            |
|----|----------------------------------------------------------|----------------------------|
| 1. | $\mathbb{F} 1, \Diamond \Box A \rightarrow \Diamond A$ ✓ | Assumption                 |
| 2. | $\mathbb{T} 1, \Diamond \Box A$ ✓                        | $\rightarrow \mathbb{F} 1$ |
| 3. | $\mathbb{F} 1, \Diamond A$                               | $\rightarrow \mathbb{F} 1$ |
| 4. | $\mathbb{T} 1.1, \Box A$                                 | $\Diamond \mathbb{T} 2$    |
| 5. | $\mathbb{T} 1, A$                                        | $\Box \mathbb{F} 4$        |
| 6. | $\mathbb{F} 1, A$                                        | $\mathbb{T} \Diamond 3$    |
|    | ⊗                                                        |                            |

## Question 8

$(\Box A \wedge \Box B) \rightarrow \Box\Box(A \rightarrow B)$  is a not theorem of KTB

|     |                                                                                  |                             |
|-----|----------------------------------------------------------------------------------|-----------------------------|
| 1.  | $\mathbb{F} 1, (\Box A \wedge \Box B) \rightarrow \Box \Box (A \rightarrow B)$ ✓ | Assumption                  |
| 2.  | $\mathbb{T} 1, \Box A \wedge \Box B$ ✓                                           | $\rightarrow \mathbb{F} 1$  |
| 3.  | $\mathbb{F} 1, \Box \Box (A \rightarrow B)$ ✓                                    | $\rightarrow \mathbb{F} 1$  |
| 4.  | $\mathbb{T} 1, \Box A$                                                           | $\wedge \mathbb{T} 2$       |
| 5.  | $\mathbb{T} 1, \Box B$                                                           | $\wedge \mathbb{T} 2$       |
| 6.  | $\mathbb{T} 1, A$                                                                | $\mathbb{T} \Box 4$         |
| 7.  | $\mathbb{T} 1, B$                                                                | $\mathbb{T} \Box 5$         |
| 8.  | $\mathbb{F} 1.1, \Box (A \rightarrow B)$ ✓                                       | $\Box \mathbb{F} 3$         |
| 9.  | $\mathbb{T} 1.1, A$                                                              | $\Box \mathbb{T} 6$         |
| 10. | $\mathbb{T} 1.1, B$                                                              | $\Box \mathbb{T} 7$         |
| 11. | $\mathbb{F} 1.1.1, A \rightarrow B$ ✓                                            | $\Box \mathbb{F} 8$         |
| 12. | $\mathbb{T} 1.1.1, A$                                                            | $\rightarrow \mathbb{F} 11$ |
| 13. | $\mathbb{F} 1.1.1, B$                                                            | $\rightarrow \mathbb{F} 11$ |

## Question 8

The tableau is open.

The countermodel takes a bit more work to describe than the previous ones.

There are three worlds,  $w_1$ ,  $w_{1.1}$  and  $w_{1.1.1}$ .

Every world can access every world *except*,  $w_1$  and  $w_{1.1.1}$  can access each other.

By the naming convention,  $w_1 R w_{1.1}$  and  $w_{1.1} R w_{1.1.1}$ ; adding reflexivity and symmetry get us the rest of the relations.

$A$  is true at all three worlds,  $B$  is true at the first and second, but not the third.



## Practice Exam

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## Question 1

$\Box(\Box p \rightarrow (q \vee r)) \vee \Box(\Box q \rightarrow p)$  in S5

|     |                                                                                       |                           |
|-----|---------------------------------------------------------------------------------------|---------------------------|
| 1.  | $\mathbb{F} 1, \Box(\Box p \rightarrow (q \vee r)) \vee \Box(\Box q \rightarrow p)$ ✓ | Assumption                |
| 2.  | $\mathbb{F} 1, \Box(\Box p \rightarrow (q \vee r))$ ✓                                 | $\vee\mathbb{F} 1$        |
| 3.  | $\mathbb{F} 1, \Box(\Box q \rightarrow p)$ ✓                                          | $\vee\mathbb{F} 1$        |
| 4.  | $\mathbb{F} 2, \Box p \rightarrow (q \vee r)$ ✓                                       | $\Box\mathbb{F} 2$        |
| 5.  | $\mathbb{T} 2, \Box p$                                                                | $\rightarrow\mathbb{F} 4$ |
| 6.  | $\mathbb{F} 2, q \vee r$ ✓                                                            | $\rightarrow\mathbb{F} 4$ |
| 7.  | $\mathbb{F} 2, q$                                                                     | $\vee\mathbb{F} 6$        |
| 8.  | $\mathbb{F} 2, r$                                                                     | $\vee\mathbb{F} 6$        |
| 9.  | $\mathbb{F} 3, \Box q \rightarrow p$                                                  | $\Box\mathbb{F} 3$        |
| 10. | $\mathbb{T} 3, \Box q$                                                                | $\rightarrow\mathbb{F} 9$ |
| 11. | $\mathbb{F} 3, p$                                                                     | $\rightarrow\mathbb{F} 9$ |
| 12. | $\mathbb{T} 3, p$                                                                     | $\Box\mathbb{T} 5$        |
|     | ⊗                                                                                     |                           |

The only branch closes, so this is **valid** in **S5**.

## Question 2

$\Diamond \Box (p \rightarrow q) \rightarrow (\Box p \rightarrow \Diamond q)$  in S5

|                                                                                                        |                                                                                                        |                            |
|--------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------------------|
| 1.                                                                                                     | $\mathbb{F} 1, \Diamond \Box (p \rightarrow q) \rightarrow (\Box p \rightarrow \Diamond q) \checkmark$ | Assumption                 |
| 2.                                                                                                     | $\mathbb{T} 1, \Diamond \Box (p \rightarrow q) \checkmark$                                             | $\rightarrow \mathbb{F} 1$ |
| 3.                                                                                                     | $\mathbb{F} 1, \Box p \rightarrow \Diamond q \checkmark$                                               | $\rightarrow \mathbb{F} 1$ |
| 4.                                                                                                     | $\mathbb{T} 1, \Box p$                                                                                 | $\rightarrow \mathbb{F} 3$ |
| 5.                                                                                                     | $\mathbb{F} 1, \Diamond q$                                                                             | $\rightarrow \mathbb{F} 3$ |
| 6.                                                                                                     | $\mathbb{T} 2, \Box (p \rightarrow q)$                                                                 | $\Diamond \mathbb{T} 2$    |
| 7.                                                                                                     | $\mathbb{T} 2, p \rightarrow q \checkmark$                                                             | $\Box \mathbb{T} 2$        |
| 8.                                                                                                     | $\mathbb{T} 2, p$                                                                                      | $\Box \mathbb{T} 4$        |
| 9.                                                                                                     | $\mathbb{F} 2, q$                                                                                      | $\Diamond \mathbb{F} 5$    |
| $  \begin{array}{c}  \swarrow \quad \searrow \\  \mathbb{F} 2, p \quad \mathbb{T} 2, q  \end{array}  $ |                                                                                                        |                            |
| 10.                                                                                                    | $  \begin{array}{cc}  \mathbb{F} 2, p & \mathbb{T} 2, q \\  \otimes & \otimes  \end{array}  $          | $\rightarrow \mathbb{T} 7$ |

Both branches close, so this is **valid** in S5.

### Question 3

$\Box(p \vee q) \rightarrow (\Diamond(\neg q \vee r) \rightarrow \Diamond(p \vee r))$  in S5

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                            |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| 1.  | $\mathbb{F} 1, \Box(p \vee q) \rightarrow (\Diamond(\neg q \vee r) \rightarrow \Diamond(p \vee r))$ ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Assumption                 |
| 2.  | $\mathbb{T} 1, \Box(p \vee q)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | $\rightarrow \mathbb{F} 1$ |
| 3.  | $\mathbb{F} 1, \Diamond(\neg q \vee r) \rightarrow \Diamond(p \vee r)$ ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $\rightarrow \mathbb{F} 1$ |
| 4.  | $\mathbb{T} 1, \Diamond(\neg q \vee r)$ ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | $\rightarrow \mathbb{F} 3$ |
| 5.  | $\mathbb{F} 1, \Diamond(p \vee r)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | $\rightarrow \mathbb{F} 3$ |
| 6.  | $\mathbb{T} 2, \neg q \vee r$ ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $\Diamond \mathbb{T} 4$    |
| 7.  | $\mathbb{T} 2, p \vee q$ ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | $\Box \mathbb{T} 1$        |
| 8.  | $\mathbb{F} 2, p \vee r$ ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | $\Diamond \mathbb{F} 5$    |
| 9.  | $\mathbb{F} 2, p$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | $\vee \mathbb{F} 8$        |
| 10. | $\mathbb{F} 2, r$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | $\vee \mathbb{F} 8$        |
| 11. | <div style="display: flex; justify-content: space-around; align-items: center;"> <div style="text-align: center;"> <math>\mathbb{T} 2, p</math><br/>  </div> <div style="text-align: center;"> <math>\mathbb{T} 2, q</math><br/> <div style="display: flex; justify-content: space-around; align-items: center;"> <div style="text-align: center;"> <math>\mathbb{T} 2, \neg q</math> ✓<br/> <math>\mathbb{F} 2, q</math><br/>  </div> <div style="text-align: center;"> <math>\mathbb{T} 2, r</math><br/>  </div> </div> </div> </div> | $\vee \mathbb{T} 7$        |
| 12. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $\vee \mathbb{T} 6$        |
| 13. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | $\neg \mathbb{T} 12$       |

## Question 4

$$(\neg \Diamond p \vee \neg \Diamond \Diamond p) \rightarrow \neg \Diamond \Diamond \Diamond p \text{ in S5}$$



|                                                                |                                                                                                           |                                                       |
|----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| 1.                                                             | $\mathbb{F} 1, (\neg\Diamond p \vee \neg\Diamond\Diamond p) \rightarrow \neg\Diamond\Diamond\Diamond p$ ✓ | Assumption                                            |
| 2.                                                             | $\mathbb{T} 1, \neg\Diamond p \vee \neg\Diamond\Diamond p$ ✓                                              | $\rightarrow\mathbb{F} 1$                             |
| 3.                                                             | $\mathbb{F} 1, \neg\Diamond\Diamond\Diamond p$ ✓                                                          | $\rightarrow\mathbb{F} 1$                             |
| 4.                                                             | $\mathbb{T} 1, \Diamond\Diamond\Diamond p$ ✓                                                              | $\neg\mathbb{F} 3$                                    |
| 5.                                                             | $\mathbb{T} 2, \Diamond\Diamond p$ ✓                                                                      | $\Diamond\mathbb{T} 4$                                |
| 6.                                                             | $\mathbb{T} 3, \Diamond p$ ✓                                                                              | $\Diamond\mathbb{T} 5$                                |
| 7.                                                             | $\mathbb{T} 4, p$                                                                                         | $\Diamond\mathbb{T} 6$                                |
| <div style="text-align: center;"> <math>\swarrow</math> </div> |                                                                                                           |                                                       |
| 8.                                                             | $\mathbb{T} 1, \neg\Diamond p$ ✓ $\mathbb{T} 1, \neg\Diamond\Diamond p$ ✓                                 | $\vee\mathbb{T} 2$                                    |
| 9.                                                             | $\mathbb{F} 1, \Diamond p$                                                                                | $\neg\mathbb{T} 8$                                    |
| 10.                                                            | $\mathbb{F} 4, p$                                                                                         | $\Diamond\mathbb{F} 9$                                |
| 11.                                                            | $\otimes$                                                                                                 | $\mathbb{F} 1, \Diamond\Diamond p$ $\neg\mathbb{T} 8$ |
| 12.                                                            |                                                                                                           | $\mathbb{F} 3, \Diamond p$ $\Diamond\mathbb{F} 11$    |
|                                                                | $\otimes$                                                                                                 |                                                       |

Both branches close, so this is **valid** in **S5**. There are many ways to close 30 / 44

## Question 5

$$(\Diamond p \wedge \Box q) \rightarrow \Diamond q \text{ in K}$$

|    |                                                                     |                            |
|----|---------------------------------------------------------------------|----------------------------|
| 1. | $\mathbb{F} 1, (\Diamond p \wedge \Box q) \rightarrow \Diamond q$ ✓ | Assumption                 |
| 2. | $\mathbb{T} 1, \Diamond p \wedge \Box q$ ✓                          | $\rightarrow \mathbb{F} 1$ |
| 3. | $\mathbb{F} 1, \Diamond q$                                          | $\rightarrow \mathbb{F} 1$ |
| 4. | $\mathbb{T} 1, \Diamond p$ ✓                                        | $\wedge \mathbb{T} 2$      |
| 5. | $\mathbb{T} 1, \Box q$                                              | $\wedge \mathbb{T} 2$      |
| 6. | $\mathbb{T} 1.1, p$                                                 | $\Diamond \mathbb{T} 4$    |
| 7. | $\mathbb{F} 1.1, q$                                                 | $\Diamond \mathbb{F} 3$    |
| 8. | $\mathbb{T} 1.1, q$                                                 | $\Box \mathbb{T} 5$        |



The trunk closes, so it is **valid** in **K**. The role of  $\Diamond p$  in this one is weird.

## Question 6

$$\Box(p \wedge \neg p) \vee \Diamond(q \vee \neg q) \text{ in K}$$

|    |                                                                   |                         |
|----|-------------------------------------------------------------------|-------------------------|
| 1. | $\models 1, \Box(p \wedge \neg p) \vee \Diamond(q \vee \neg q)$ ✓ | Assumption              |
| 2. | $\models 1, \Box(p \wedge \neg p)$ ✓                              | $\vee\mathcal{E}$ 1     |
| 3. | $\models 1, \Diamond(q \vee \neg q)$                              | $\vee\mathcal{E}$ 1     |
| 4. | $\models 1.1, p \wedge \neg p$                                    | $\Box\mathcal{E}$ 2     |
| 5. | $\models 1.1, q \vee \neg q$ ✓                                    | $\Diamond\mathcal{E}$ 3 |
| 6. | $\models 1.1, q$                                                  | $\wedge\mathcal{E}$ 5   |
| 7. | $\models 1.1, \neg q$ ✓                                           | $\vee\mathcal{E}$ 5     |
| 8. | $\top 1.1, q$                                                     | $\neg\mathcal{E}$ 7     |



This is also valid, and it's also wrong for the same reason.

## Question 6

Not at all on the test digression.

This example shows that **K** has a very weird property. It is not **Halldén-complete**.

A logician, Sören Halldén, in the 1950s noticed that some logics have the odd property that  $X \vee Y$  is a theorem even though (a) neither  $X$  nor  $Y$  is, and (b)  $X$  and  $Y$  have no variables in common.

We've come to call such logics Halldén-incomplete, and the opposite property Halldén-completeness.

In general, the logics that are interesting and get used in philosophical applications are Halldén-complete.

## Question 7

$$\Box \neg (p \vee \neg q) \rightarrow \Box p \text{ in K}$$

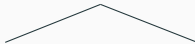
|    |                                                               |                            |
|----|---------------------------------------------------------------|----------------------------|
| 1. | $\mathbb{F} 1, \Box \neg(p \vee \neg q) \rightarrow \Box p$ ✓ | Assumption                 |
| 2. | $\mathbb{T} 1, \Box \neg(p \vee \neg q)$                      | $\rightarrow \mathbb{T} 1$ |
| 3. | $\mathbb{F} 1, \Box p$ ✓                                      | $\rightarrow \mathbb{F} 1$ |
| 4. | $\mathbb{F} 1.1, p$                                           | $\Box \mathbb{F} 3$        |
| 5. | $\mathbb{T} 1.1, \neg(p \vee \neg q)$ ✓                       | $\Box \mathbb{T} 2$        |
| 6. | $\mathbb{F} 1.1, p \vee \neg q$ ✓                             | $\neg \mathbb{T} 5$        |
| 7. | $\mathbb{F} 1.1, p$                                           | $\vee \mathbb{F} 6$        |
| 8. | $\mathbb{F} 1.1, \neg q$ ✓                                    | $\vee \mathbb{F} 6$        |
| 9. | $\mathbb{T} 1.1, q$                                           | $\neg \mathbb{F} 8$        |

The tree is open, so this is not a theorem, it is **invalid** in **K**. The countermodel has two worlds:  $w_1$  and  $w_1.1$ , with  $w_1 R w_{1.1}$ . At  $w_{1.1}$ ,  $p$  is false and  $q$  is true.



## Question 8

$$\Box(p \rightarrow (q \wedge r)) \rightarrow (\Diamond\Diamond p \rightarrow \Diamond(q \vee s)) \text{ in K}$$

|                                                                                                                            |                                                                                                                    |                             |
|----------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|-----------------------------|
| 1.                                                                                                                         | $\mathbb{F} 1, \Box(p \rightarrow (q \wedge r)) \rightarrow (\Diamond\Diamond p \rightarrow \Diamond(q \vee s))$ ✓ | Assumption                  |
| 2.                                                                                                                         | $\mathbb{T} 1, \Box(p \rightarrow (q \wedge r))$                                                                   | $\rightarrow \mathbb{F} 1$  |
| 3.                                                                                                                         | $\mathbb{F} 1, \Diamond\Diamond p \rightarrow \Diamond(q \vee s)$ ✓                                                | $\rightarrow \mathbb{F} 1$  |
| 4.                                                                                                                         | $\mathbb{T} 1, \Diamond\Diamond p$ ✓                                                                               | $\rightarrow \mathbb{F} 3$  |
| 5.                                                                                                                         | $\mathbb{F} 1, \Diamond(q \vee s)$                                                                                 | $\rightarrow \mathbb{F} 3$  |
| 6.                                                                                                                         | $\mathbb{T} 1.1, \Diamond p$ ✓                                                                                     | $\Diamond \mathbb{T} 4$     |
| 7.                                                                                                                         | $\mathbb{T} 1.1, \Box(p \rightarrow (q \wedge r))$                                                                 | $4 \Box 2$                  |
| 8.                                                                                                                         | $\mathbb{F} 1.1, \Diamond(q \vee s)$                                                                               | $4 \Diamond 5$              |
| 9.                                                                                                                         | $\mathbb{T} 1.1.1, p$                                                                                              | $\Diamond \mathbb{T} 6$     |
| 10.                                                                                                                        | $\mathbb{T} 1.1.1, p \rightarrow (q \wedge r)$                                                                     | $\Box \mathbb{T} 7$         |
| <div style="text-align: center;">  </div> |                                                                                                                    |                             |
| 11.                                                                                                                        | $\mathbb{F} 1.1.1, p$                                                                                              | $\rightarrow \mathbb{T} 10$ |
| 12.                                                                                                                        | $\otimes$ $\mathbb{T} 1.1.1, q$ ✓                                                                                  | $\wedge \mathbb{T} 11$      |
| 13.                                                                                                                        | $\mathbb{T} 1.1.1, r$                                                                                              | $\wedge \mathbb{T} 11$      |
| 14.                                                                                                                        | $\mathbb{F} 1.1.1, q \vee s$                                                                                       | $\Diamond \mathbb{F} 8$     |
| 15.                                                                                                                        | $\mathbb{F} 1.1.1, q$                                                                                              | $\vee \mathbb{F} 14$        |

## Question 9

$$(\Diamond \Box p \wedge \Diamond \Box q) \rightarrow \Diamond(p \wedge q) \text{ in K}$$

|    |                                                                 |                            |
|----|-----------------------------------------------------------------|----------------------------|
| 1. | $\mathbb{F} 1, (p \wedge \Diamond \Box p) \rightarrow \Box p$ ✓ | Assumption                 |
| 2. | $\mathbb{T} 1, p \wedge \Diamond \Box p$ ✓                      | $\rightarrow \mathbb{F} 1$ |
| 3. | $\mathbb{F} 1, \Box p$ ✓                                        | $\rightarrow \mathbb{F} 1$ |
| 4. | $\mathbb{T} 1, p$                                               | $\wedge \mathbb{T} 2$      |
| 5. | $\mathbb{T} 1, \Diamond \Box p$ ✓                               | $\wedge \mathbb{T} 2$      |
| 6. | $\mathbb{F} 1.1, p$                                             | $\Box \mathbb{F} 3$        |
| 7. | $\mathbb{T} 1.2, \Box p$                                        | $\Diamond \mathbb{T} 5$    |
| 8. | $\mathbb{T} 1.2, p$                                             | $\mathbb{T} \Box 7$        |

## Question 9

This is open, so it isn't valid.

The countermodel has three worlds,  $w_1$ ,  $w_{1.1}$  and  $w_{1.2}$ .

$w_1$  can access everything, the other two can only access themselves. And  $p$  is true at  $w_1$  and  $w_{1.2}$ , but false at  $w_{1.1}$ .

## Question 10

$$\Box\Diamond\Box\Diamond p \rightarrow \Box\Diamond p \text{ in K}$$

|    |                                                                             |                            |
|----|-----------------------------------------------------------------------------|----------------------------|
| 1. | $\mathbb{F} 1, \Diamond \Box \Diamond \Box p \rightarrow \Diamond \Box p$ ✓ | Assumption                 |
| 2. | $\mathbb{T} 1, \Diamond \Box \Diamond \Box p$ ✓                             | $\rightarrow \mathbb{F} 1$ |
| 3. | $\mathbb{F} 1, \Diamond \Box p$                                             | $\rightarrow \mathbb{F} 1$ |
| 4. | $\mathbb{T} 1.1, \Box \Diamond \Box p$                                      | $\Diamond \mathbb{T} 2$    |
| 5. | $\mathbb{F} 1.1, \Diamond \Box p$                                           | $4 \Diamond 3$             |
| 6. | $\mathbb{T} 1.1, \Diamond \Box p$                                           | $\mathbb{T} \Box 4$        |



The trunk closes, so this is **valid**. There are *a lot* of ways to do this one; this is one of the more direct.